WESTMINSTER
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System Analysis & Design

Lecture 3

Agenda

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1. Last week recap
2. Prototyping methodology

System Develop Methodologies

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Structural:

- Waterfall
- Structured Software Development Methodology (SSDM)

Iterative and Incremental:

- RAD
- Spiral
- eXtreme Programming
- Scrum
- Etc.

Prototyping:

- Prototyping

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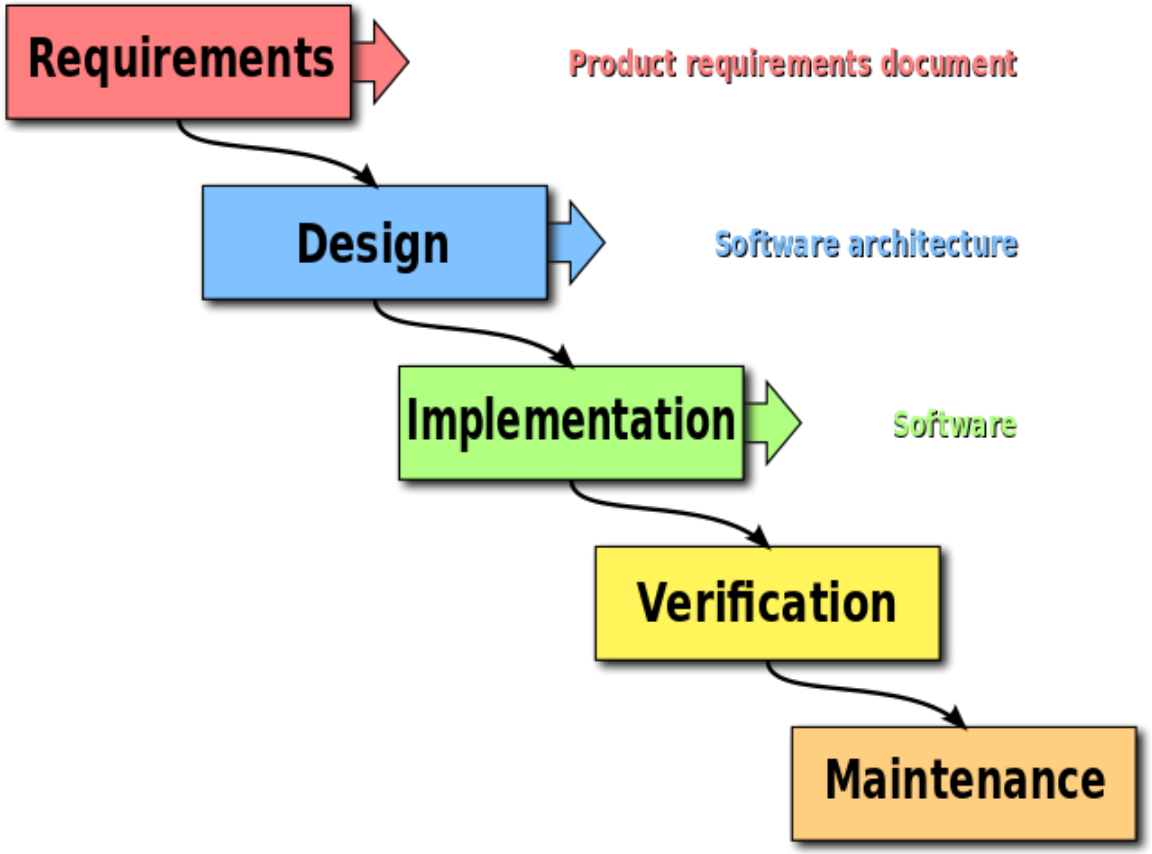
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Recap

What did we discuss last week?

Waterfall Methodology

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Model Steps

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1. **Requirements** – Documents all requirements.
2. **Design** – Based on the design, build a detailed system design.
3. **Implementation** – Build the system based on the design.
4. **Verification** – Test the new system and fix all the bugs.
5. **Maintenance** – Deploy and maintain the system.

Methodology Features

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- Steps are carried out in fixed order (sequencial).
- Each step must be fully finished before moving to the next (structured).
- Input of a step is the output of the previous step.
- Each step is documented. The output of each step is a documented.

Problems it solves

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What problems does it address?

- Changing project later on is costly. Tens or hundreds of times costly. That's why we need to spend time and resources to get them right at very first step.
- People may leave your team. They may be the carriers of the project related knowledge. That's why you need to document everything in a standard way.
- Gives concrete steps, that are easy to understand. Everyone knows at what phase the project is, and is the main goal at the moment – synergy.

When to use it?

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- Projects that have clearly defined final product.
- Stable and clear requirements i.e. requirements do not change.
- Clearly defined and fixed scope.
- Large scale project. For small projects it is not as effective.



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Prototyping

Prototyping as a software development methodology.

System Develop Methodologies

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Prototyping:

- Prototyping

First point and click prototypes

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Tale of Drew Houston & Dropbox

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BlackRock Is Leading the Deal Alongside Previous Backers

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Prototyping

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- So far, we've seen how prototyping process is used in product development (manufacturing and business).

- Now, let's see how we can use the prototyping idea as a method of developing a software product.

Software Prototyping Process

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- Software prototyping is the process of creating an incomplete model of the future full-featured software.

- **The process:**
  - Identify basic requirements
  - Develop Initial prototype
  - Review with end users
  - Revise and enhance

Types of Software Prototypes

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- **Throwaway prototype** – process of creating a rapid model (demonstration or marketing).
- **Evolutionary prototype** – build a system then refine.
- **Incremental** – final product built as separate prototypes.

Types of Software Prototypes

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Prototyping Approaches

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- The objective of ***evolutionary prototyping*** is to deliver a working system to end-users

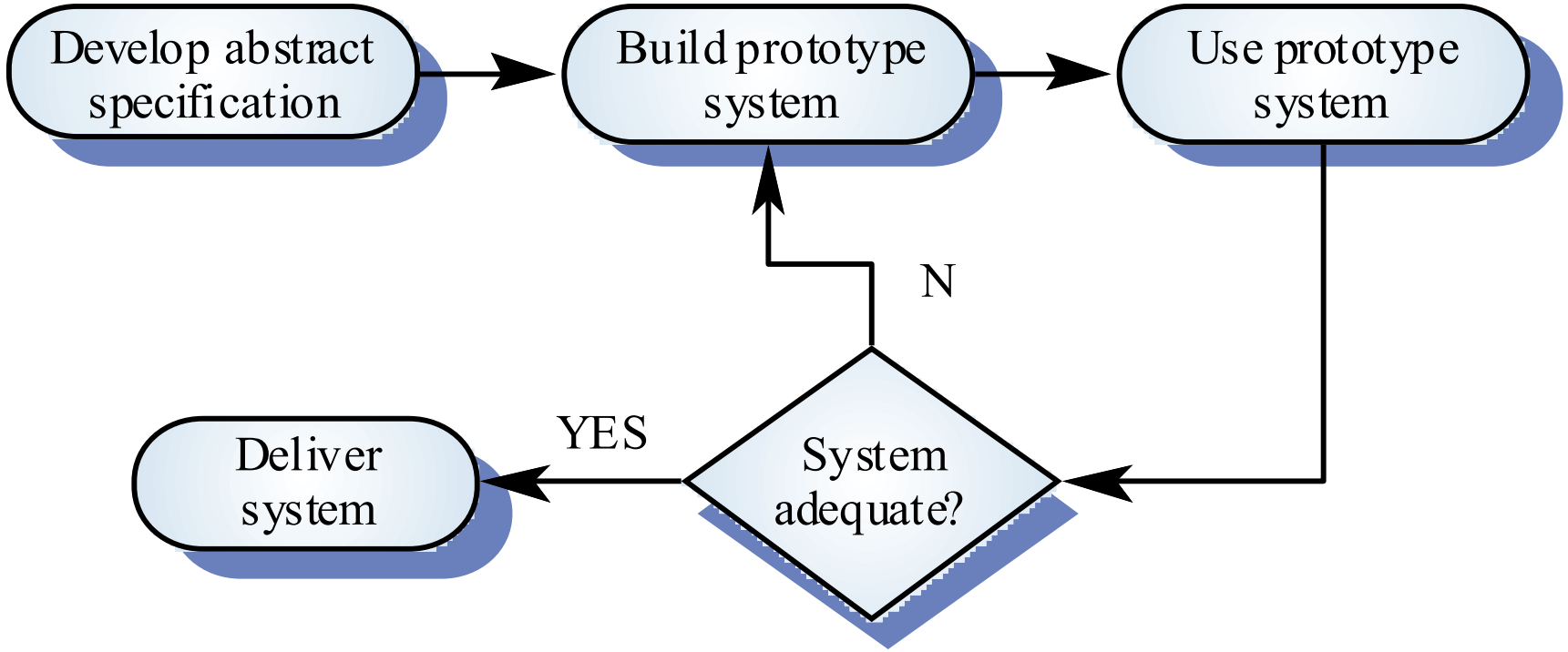
  The development starts with those requirements which are best understood.

- The objective of ***throw-away prototyping*** is to validate or derive the system requirements

  The prototyping process starts with those requirements which are poorly understood.

Evolutionary Prototyping

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Evolutionary Prototyping

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- Must be used for systems where the requirements specification cannot be developed in advance.
- Based on techniques which allow rapid system iterations.
- Verification is impossible as there is no specification.
- Validation means demonstrating the adequacy of the system.

Evolutionary Prototyping

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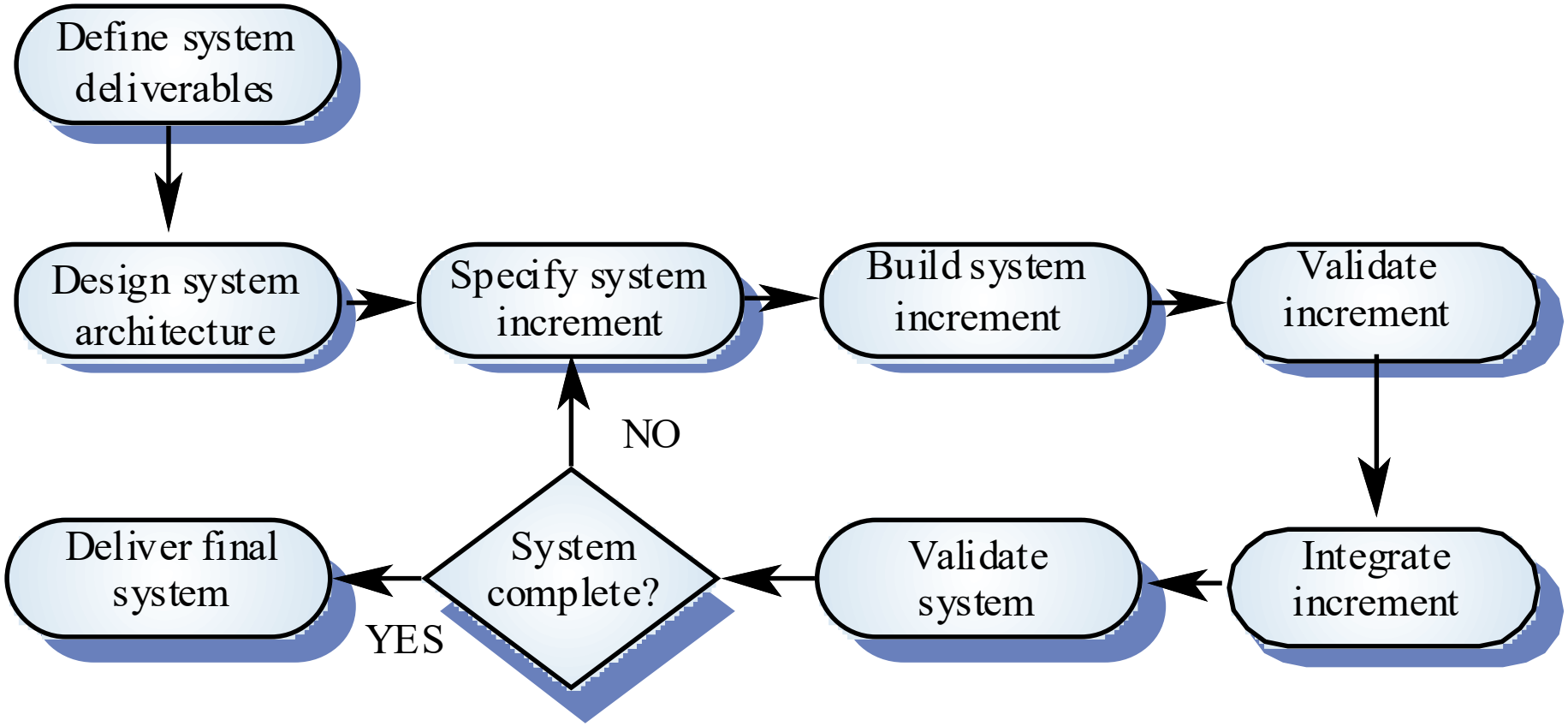
Evolutionary Prototyping

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- Advantages
- Challenges
- Technologies

Incremental Prototyping

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Incremental Prototyping

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- System is developed and delivered in increments after establishing an overall architecture.
- Requirements and specifications for each increment may be developed.
- Users may experiment with delivered increments while others are being developed.
- Intended to combine some of the advantages of prototyping:
  - More manageable process.
  - Better system structure.

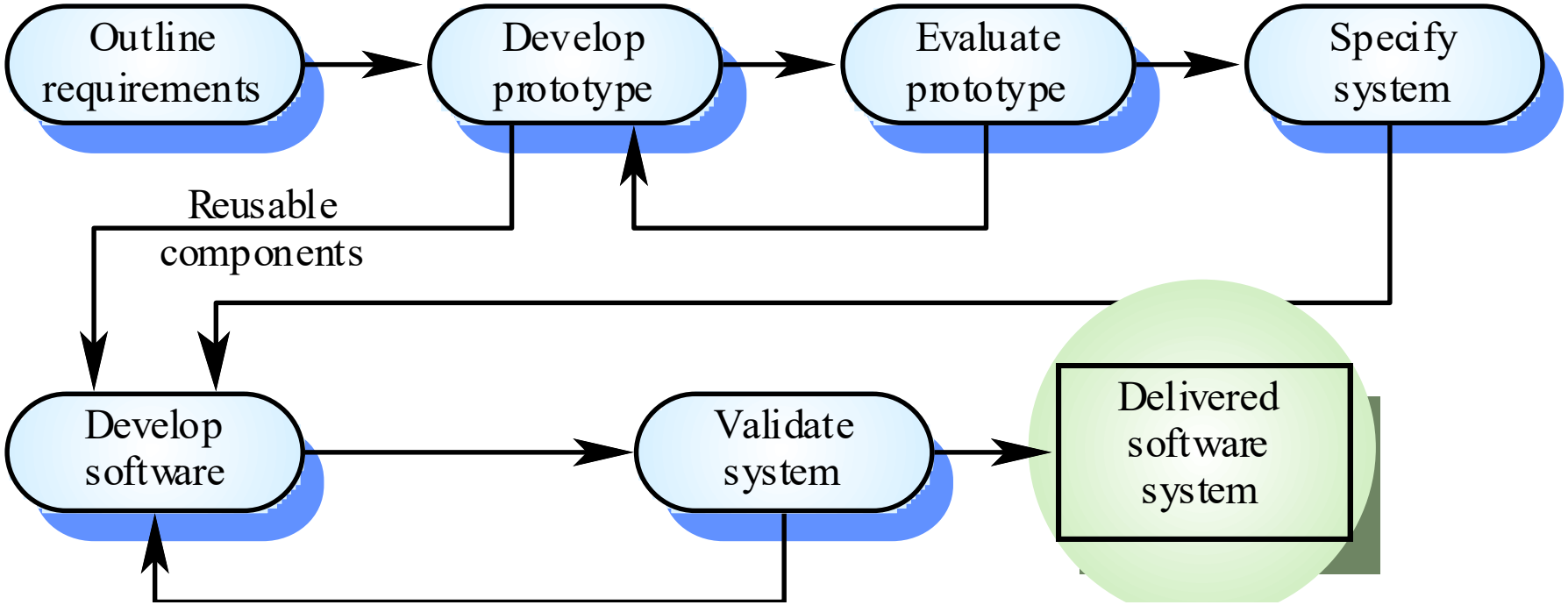
Incremental Prototyping

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- Advantages
- Challenges
- Technologies

Throw-away Prototyping

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Throw-away Prototyping

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- Used to reduce requirements risk.
- The prototype is developed from an initial specification, delivered for experiment then discarded.
- The throw-away prototype must NOT be considered as a final system.
  - Some system characteristics may have been left out.
  - There is no specification for long-term maintenance.
  - The system will be poorly structured and difficult to maintain.

Throw-away Prototyping

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- Advantages
- Challenges
- Technologies

Prototyping Methodology

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- You might have noticed that Prototyping has the iterative process like Agile methodologies.



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Questions?


The End